

Three explicit physics versions

Numerical near-rigid assumption only. No material calibration or manipulation-success claim.

LEGACY: solref [0.02, 1]; solimp [0.9, 0.95, 0.001, 0.5, 2]. Historical reference only.

IMP99: solref [0.02, 1]; solimp [0.99, 0.99, 0.001, 0.5, 2].

TC10_IMP99: solref [0.01, 1]; identical constant-0.99 solimp.

All candidates: runtime friction [0.5, 0.5, 0.01, 0.003, 0.003]; effective dim 6.

Production: dt 2 ms; Newton / 100 iterations / tolerance 1e-8; Euler; elliptic cone; impratio 10.

31 explicit hand-object pairs. Native geometry, gains, tendons and unrelated contacts preserved.

Decision: NO FREEZE. Actual-hand simultaneous M/R/L validation was not established.